An Audited Evaluation of Event-Aware Retrieval for 24-Hour PM2.5 Forecasting

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***Abstract*—Retrieval augmentation gives forecasting models access to historical examples, but its value can be overstated when the memory is sparse, the comparison baseline is weak, or contextual channels are not tested separately. This study audits Event-TimeRAF, which consists of a Source-Audited Context Builder, a Leakage-Safe Event-Context Retriever, and a Drift-Aware Forecast and Evidence Head. EPA AirData PM2.5, NOAA Global Hourly weather, calendar variables, and hash-verified NOAA Storm Events records are aligned for Los Angeles County from 2019 to 2024. The evaluation uses a 168-hour lookback, a 24-hour horizon, 7,199 test origins, four knowledge-base strides, event-weight controls, retrieval placebos, and overlap-aware paired inference. The best model is the weather-calendar XGBoost baseline, with MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379. Full Event-TimeRAF obtains MSE 26.734, and its difference from the context baseline is unresolved. Frozen Chronos-Bolt obtains MSE 28.941. Retrieval fusion obtains 28.733, while a climatology fusion control obtains 27.450 and significantly outperforms it. A denser six-hour memory improves retrieval-only MSE to 34.616, but the feature model still remains behind the context baseline. An event weight of 0.2 changes 91.4% of test top-k sets but worsens retrieval error. The tested event representation therefore adds no forecast value. Since Storm Events has no machine-readable publication timestamps, event-conditioned findings remain retrospective sensitivity results.**

***Index Terms*—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, event context, distribution-shift diagnostics, forecast traceability.**

I. Introduction

Fine particulate matter (PM2.5) is one of the key urban air quality risks, due to its relation to health effects on respiration and cardiovascular system [1]. Correct short term PM2.5 prediction can be used for the purpose of public health warning, exposure avoidance, transportation scheduling and environmental management. However, the dynamics of PM2.5 is challenging to model based on pollutant history in Los Angeles County. Concentrations may vary depending on wind speed, boundary layer behavior, rain events, season, holidays, wildfire smoke and/or on unusual storm or high wind events. These factors means that the forecasting problem is time and context dependent.

Recent time-series foundation models (TSFMs) try to enhance the generalization ability of forecasting by learning from a huge amount of time-series data and transferring the forecasting capability to a new domain. Some models are available, such as TimesFM, Moirai, MOMENT, Chronos and Tiny Time Mixers [2]–[6]. Concurrently, it has been demonstrated that non-parametric memory can be useful for foundation models to utilize a relevant external knowledge at inference time [7], [8] through retrieval-augmented generation. In order to merge these two ideas for zero-shot TSF, TimeRAF retrieves relevant historical sequences from a curated knowledge base and use Channel Prompting [9] to inject them to a frozen TSFM.

The domain-specific retrieval question is not fully answered by numerical time-series retrieval alone. A historical PM2.5 window can be a similar window to the current one, but can be related to different weather or event conditions. On the other hand, a window having moderate numerical similarity might be more informative if having high wind, rain, holiday, smoke or stagnation context. This encourages forecasting to be done within a retrieval system and to consider source records, weather, calendar data and distribution-shift indicators as components of the forecasting evidence instead of as metadata.

In this study, Event-TimeRAF is evaluated as an event-aware retrieval framework for PM2.5 forecasting under distribution shift. The method takes the idea of retrieval from TimeRAF and adapts it to an official-source environmental setting, but it does not duplicate TimeRAF's learned retriever or Channel Prompting mechanism. Event-TimeRAF has three named parts: a Source-Audited Context Builder (SACB), a Leakage-Safe Event-Context Retriever (LSER), and a Drift-Aware Forecast and Evidence Head (DFEH). Together they examine how official records can be prepared for retrieval, how historical candidates can be selected without temporal leakage, and whether retrieval adds value beyond a strong weather-calendar baseline.

The main contributions are listed below:

- A reproducible mechanism, called the Source-Audited Context Builder, that maintains consistency between EPA PM2.5, NOAA weather, calendar, and hash-verified NOAA Storm Events records, and maintains source identity, coverage checks, and the assumption of event-availability.
- *Leakage-Safe Event-Context Retriever:* a historical analogue mechanism that applies a query–candidate embargo and is tested across memory density, neighbor count, event weight, and event-stratified controls.
- *Drift-Aware Forecast and Evidence Head:* an evaluation layer that compares direct XGBoost, frozen Chronos-Bolt, retrieval fusion, and non-retrieval fusion controls while retaining shift and provenance evidence.

The evaluation is conservative. It only reports manifest-backed comparisons, uses 2,000-resample paired block intervals and overlap-aware Diebold–Mariano tests, and does not interpret event association as causation. The negative findings separate a working and auditable retrieval pipeline from evidence that retrieval improves forecasting.

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II. Literature Review and Related Work

A. Statistical, Machine-Learning, and Deep Forecasters

Forecasting techniques differ in their inductive biases, data requirements and computational costs. Autoregressive and seasonal models are informative baselines because their failure modes are readily inspected. Tree ensembles (e.g., XGBoost and LightGBM) build on this configuration by supporting nonlinear interactions among lag, rolling, weather and calendar features without requiring a sequence encoder [10], [11]. Recurrent LSTM and GRU models, however, learn stateful temporal representations [12], [13]. Reproducible comparison across heterogeneous time-series domains is also supported by large benchmark archives such as the Monash Forecasting Repository [14]. Along this line, transformer families model longer dependencies: Informer reduces attention cost; Autoformer and FEDformer decompose time series; Crossformer and iTransformer model cross-variable structure; and TimesNet rearranges temporal variation into two-dimensional patterns [15]–[20]. These benefits do not necessarily mean that a more complex architecture is always required. Simple decomposition- or patch-based inductive biases remain competitive in DLinear and PatchTST [21], [22], while TimeMixer explicitly decomposes and mixes information at multiple scales [23]. Thus, Event-TimeRAF does not assume a deep backbone to be the default winner, but retains seasonal and boosted-tree baselines.

B. Time-Series Foundation Models

Time-series foundation models (TSFMs) aim to transfer across datasets as opposed to training a new representation from scratch for every time series. The former is a decoder-only forecasting design (TimesFM), the latter a universal forecasting Transformer (Moirai), the next one supports multiple time-series tasks (MOMENT), the next one encodes values as tokens (Chronos), and the last one focuses on compact zero- and few-shot transfer (Tiny Time Mixers) [2]–[6]. Timer, Timer-XL, UniTime, and Time-LLM explore generative pre-training, long-context scaling, cross-domain unification, or language-model reprogramming [24]–[27]. It was also found that frozen pretrained models can be used as general computation engines for analyzing time series [28]. All these studies encourage zero-shot evaluation, and also indicate that the performance of the transfer also relies on input representation and domain shift. Therefore, in the present study, we do not want to assume that the models based on a foundation model are better, and we want to compare frozen Chronos-Bolt and locally supervised models on the same forecast origins.

C. Retrieval-Augmented Forecasting

Retrieval augmentation separates parametric knowledge from an external memory. The retrieve-then-condition pattern was pioneered by RAG and Dense Passage Retrieval in NLP [7], [8]. In the context of time series, retrieval has been investigated in diffusion forecasting and assisted-generation workflows [29], [30]. TimeRAF is the most similar method: it builds a time-series knowledge base, uses an MLP dual-encoder retriever, finds the top k candidates and incorporates candidate embeddings into a frozen TSFM through Channel Prompting [9]. Its ablations examine retrieval quality, integration, the number of candidates, knowledge-base composition and backbone selection.

Event-TimeRAF is based on the same external-memory principle but is designed to answer a different experimental question. It uses transparent random, cosine and event-context similarity instead of a learned dual encoder, and feature- or forecast-level fusion instead of internal Channel Prompting. The current implementation is therefore an audited environmental adaptation intended to determine whether contextual historical analogues assist a local PM2.5 forecasting task, rather than a reproduction of TimeRAF.

D. Air-Quality Forecasting and Official Context

History of pollutants along with temperature, humidity, wind, precipitation and pressure are commonly used in PM2.5 forecasts. In recent years, there have been reviews of single-station recurrent models to attention-based, hybrid, and spatiotemporal models [31], [32]. For example, AirFormer models at a nationwide level by separating the deterministic spatiotemporal modeling from the stochastic uncertainty estimation [33]. These multi-station designs account for spatial dependence, while the current study considers a county-level aggregate, and a source of contextual evidence. Observed PM2.5 data is provided by the EPA AirData, hourly meteorology is provided by NOAA Integrated Surface Database, and structured hazard data is provided by NOAA Storm Events [34]–[36]. Evidence of smoke could be included in NOAA Hazard Mapping System products, but were not used in the run reported [37].

E. Distribution Shift and Forecast Explanations

Temporal nonstationarity may lead to changes in means, variances and relationships between inputs and future values. Adaptive normalization methods address this problem by estimating statistics at local temporal scales [38]; Event-TimeRAF instead uses interpretable indicators of shift to audit regimes without claiming that shift has been eliminated. There are also varying degrees of scope of explanation. SHAP can be used to attribute predictions to model inputs [39] and information-theoretic saliency can be used for counterfactual inspection of probabilistic forecasts [40]. The explanation layer implemented is more restricted: it only captures XGBoost signed effects, retrieved cases, event identifiers, drift components, and a diagnostic uncertainty scale. These records help with the traceability, however, they do not help to infer causal relationships between weather and events.

III. Critical Gaps and Limitations of Previous Studies

Existing studies leave three gaps that need to be addressed in the context of event sensitive urban air-quality forecasting. First, temporal forecasters and temporal numerical retrieval systems do not necessarily take care of the provenance, coverage, and availability assumptions of local contextual records. A numerically similar PM2.5 window may happen during a weather regime or event different from that indicated by the retrospective event label, and a retrospective event label could be misinterpreted as information at forecast time. The SACB aims to fill this void by incorporating official records through explicit source, coverage and availability audits.

Second, historical retrieval may include look-ahead bias if there is overlap between the candidate input and the query context, or between the candidate target and the query context. The fact that it is similar doesn't mean that a near duplicate future doesn't get into the retrieved evidence. The LSER resolves this with its training-history restriction and strict temporal embargo, followed by PM2.5, weather, calendar and event similarity terms, which can be audited independently.

Third, retrieval gain is sometimes reported without evaluation of whether there is already a powerful local context model that captures the useful signal, whether retrieval gain still exists under distribution shift, and whether there is evidence to explain retrieval gain. The DFEH fills this gap by conducting assessments of both the direct XGBoost and frozen Chronos-Bolt paths on common origins, highlighting shift indicators, and maintaining machine-readable feature, retrieval, event and uncertainty evidence. This is a component of comparison and auditability; causal explanations are not produced.

These mappings form the boundaries of the novelty of the study. Event-TimeRAF does not have a learned dual-encoder retriever or internal Channel Prompting, and the current experiment is only done on a single county, without establishing geographic generalization.

IV. Core Contributions

The proposed framework provides three mechanisms, that correspond one on one with the three gaps listed above:

- *Source-Audited Context Builder (SACB):* builds an aligned context that is 85 dimensions based on official PM2.5 and meteorological data, calendar and event information, and retains source and availability metadata.
- *Leakage-Safe Event-Context Retriever (LSER):* retrieves training-history analogues only after implementing a candidate-target embargo and provides exposure of the contribution of each similarity channel and retrieved trajectory evidence.
- *Drift-Aware Forecast and Evidence Head (DFEH):* integrates all of the above into a single chronological evaluation contract.

The empirical contribution is intentionally qualified: weather-calendar context produces the clearest reduction in error, denser retrieval improves only the retrieval-only path, and the tested event channel does not improve forecast accuracy.

V. Problem Formulation

Let x_t be the measurement of PM2.5 at time t. The lookback window for the air quality is defined by Equation (1) for forecast time t.

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

Here, L is the lookback length, W_t contains weather covariates aligned with the lookback window, C_t contains calendar features, and E_t contains event-context variables available at the forecast origin under the recorded source assumption. The target in Equation (2) corresponds to the input in Equation (1).

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where H represents the horizon of forecast. Equation (2) thus represents the 24-step vector for each allowable origin.

The forecasting function is given by Equation (3):

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is a retrieved time-series knowledge, R_t^{ev} is an event knowledge retrieved and D_t is a concept-drift indicator set. The target setting which has been implemented is: L = 168 hours, H = 24 hours. The full information interface is given in Equation (3) and the following section describes the modules that are implemented to build the arguments.

The time based split is given by Equation (4).

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

Here the first 70% of observations are employed to train the system, the second 15% for validation and the final 15% for testing. The order in (4) is preserved for feature fitting, retrieval eligibility, model selection and final evaluation.

The notation used in the rest of the report is summarized in Table I.

TABLE I
Main Symbols Used in the Proposed Formulation

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
q_t	SACB context vector in $\mathbb{R}^{85}$
G_t	LSER top-k retrieved set
$\rho(G_t)$	Retrieval summary in $\mathbb{R}^{51}$
$\mathcal{O}_t$	Machine-readable DFEH evidence record
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

VI. Methodology

The Event-TimeRAF is composed of three main components. The Source-Audited Context Builder (SACB) provides a way to translate from the heterogeneous official records to a chronological hourly context while preserving the provenance and availability metadata. The Leakage-Safe Event-Context Retriever (LSER) builds the embargoed historical memory as well as retrieves the analogues based on PM2.5 similarity, weather similarity, calendar similarity and event similarity. The Drift-Aware Forecast and Evidence Head (DFEH) reveals feature level and forecast level prediction paths, prediction shift indicators, and a machine readable explanation record. No image operator, learned attention block, or internal TimeRAF Channel Prompting is implied with these components which are deterministic or supervised modules over one dimensional time-series and tabular inputs.

A. Framework Overview and Forward Pass

The whole dataflow is summarized in figure 1. The first data to enter the SACB are EPA PM2.5, NOAA weather, calendar and NOAA Storm Events records that are audited and aligned then transformed into a query context and input-target windows. The LSER filters the knowledge base of training-histories, ranks candidates eligible for the filter and returns the retrieved candidate futures and retrieval statistics. The DFEH then writes those statistics to 24 direct regressors of XGBoost, evaluates the separate frozen fusion path of Chronos-Bolt, calculates drift evidence and writes the forecast and supporting record of the forecast.

The aligned Event-TimeRAF context is mapped to a forecast, via Equation (5), for each forecast origin t.

$$\hat{Y}_{t+1:t+H} = F_\theta\big(\phi(X_t, W_t, C_t, E_t), G_t, D_t\big), \tag{5}$$

In which, phi(.) is the tabular feature construction, G_t is the retrieved historical evidence and D_t is the drift evidence. A major limitation of the method is related to the availability of all the components of G_t and D_t at or before the time of the forecast. The tensor level forward pass below is the expanded form of Equation (5).

In tensor notation, Equation (6) expands the forward pass that was implemented.

$$\begin{aligned}
Q &= \mathrm{SACB}(X, W, C, E) && \in \mathbb{R}^{B\times 85},\\
G &= \mathrm{LSER}(Q, \mathcal{K}) && \in \mathbb{R}^{B\times 8\times 24},\\
U_h &= [Q \,\|\, \rho(G) \,\|\, d \,\|\, C_h^+] && \in \mathbb{R}^{B\times 151},\\
\hat{Y}^X &= [f_1(U_1), \ldots, f_{24}(U_{24})] && \in \mathbb{R}^{B\times 24},\\
(\hat{Y}, \mathcal{O}) &= \mathrm{DFEH}(\hat{Y}^X, \hat{Y}^C, \hat{Y}^R, d, G) && \in \mathbb{R}^{B\times 24}\times\mathcal{E},
\end{aligned} \tag{6}$$

where Q is the origin context (85-dimensional), $\rho(G) \in \mathbb{R}^{B\times 51}$ contains the retrieved trajectory, spread, two similarity summaries and the number of candidates, $d \in \mathbb{R}^{B\times 6}$ are five drift components and their mean score, and $C_h^+ \in \mathbb{R}^{B\times 9}$ is the known calendar vector for the horizon h. The symbol E denotes the schema of the evidence record O. Equation (6) describes both reported forecasting paths without presenting the XGBoost path as part of the frozen foundation model.

B. Source-Audited Context Builder

The motivation behind the SACB is to maintain the data provenance and temporal availability prior to model fitting. Figure 2 illustrates that source specific checks are done first and then the feature is built. EPA monitor records are aggregated to an hourly county median, NOAA weather data from the selected surface station are matched up to the same time, event information is hash verified against an event manifest and calendar information is calculated from the local time. A windowed and chronologically split context is thus generated, no target value is ever interpolated.

1) Dataset Construction: The hourly observations of PM2.5 are obtained from the US EPA/AirData/AQS parameter 88101 (Los Angeles County) [34]. The data audit chooses one monitoring site based on coverage and continuity criteria, if there is one monitoring site that meets the coverage criterion. Since no particular monitor met this criterion, the target is the county median measured by the official Los Angeles County AQS monitors for every hour. Times are to the `America/Los_Angeles` time zone, daylight saving time is handled explicitly. Targets are not interpolated, short input gaps can only be filled with past observations. Weather covariates are obtained from the closest weather station from the NOAA Integrated Surface Database (ISD) [35] that is suitable. Calendar elements include cyclic hour, weekday and month terms, weekend, US federal holiday and California holidays.

The feature groups are merged at all the origins using Equation (7).

$$q_t = [p_t \,\|\, w_t \,\|\, c_t \,\|\, e_t] \in \mathbb{R}^{23+14+9+39} = \mathbb{R}^{85}, \tag{7}$$

Here the feature vectors p_t, w_t, c_t, and e_t are the PM2.5 feature vector, weather feature vector, calendar feature vector and event feature vector. The PM2.5 group consists of fixed lags, rolling moments and extrema, differences, the current value and a fill flag. The weather group contains six measured variables, a precipitation-missing flag, derived condition flags, a stagnation proxy, and 24-hour running means. Event group consists of group of 1 hour counts, group of active counts, 24/72/168 hour counts, category specific 72 hour counts and event-burst ratio. Therefore, Equation (7) provides the ordering of features as well as the 85 dimensional SACB output to be fed to downstream modules.

The statistics for normalization are only fit on the training split. Both query and candidate windows are normalized in the same way, based on the "TimeRAF" principle that sequences retrieved and input must follow the same way of normalization. The reported run contains 48,332 valid windows with $X \in \mathbb{R}^{48332\times 168}$ and $Y \in \mathbb{R}^{48332\times 24}$.

C. Leakage-Safe Event-Context Retriever

The LSER differentiates between temporal eligibility and relevance. First, the embargo is shown (Figure 3): if the entire target of the embargo is before the start of the query lookback then the candidate is considered. The selected candidates are then compared on four similarity channels which are independently stored. Selected futures are on the same scale as the query, and summarized for the forecasting head. This ordering is to ensure that a high similarity score does not overrule the temporal constraint.

1) Time-Series Knowledge Base: Historical PM2.5 windows are stored in the time-series knowledge base together with their identifiers, times, input and future values, weather and calendar summaries, normalized vectors, split, and normalization identifier. The primary configuration uses a 24-hour stride and contains 1,521 training-history candidates. Candidate windows may overlap one another, but Equation (9) prevents overlap between a candidate target and the query lookback. Sensitivity runs use strides of 192, 24, 6, and 1 hours, producing 191, 1,521, 5,644, and 34,020 candidates.

Equations (8) formally defines the knowledge base.

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^{N}, \tag{8}$$

In this case, u_i are the list of candidate origins, X_i and Y_i are the candidate lookback and future windows, m_i the list of

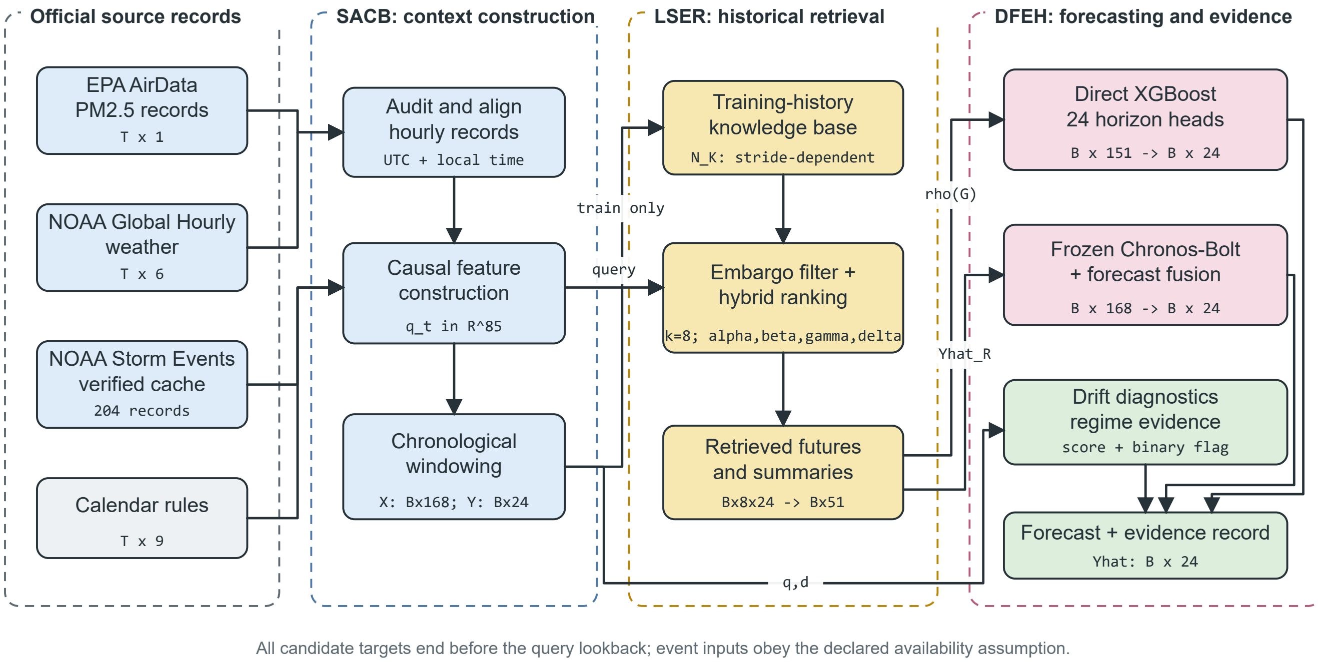

Fig. 1. Proposed Event-TimeRAF pipeline. Official records pass through the Source-Audited Context Builder (SACB), training-history candidates are selected by the Leakage-Safe Event-Context Retriever (LSER), and the Drift-Aware Forecast and Evidence Head (DFEH) produces 24-hour forecasts and traceable evidence. Tensor annotations show the reported configuration.

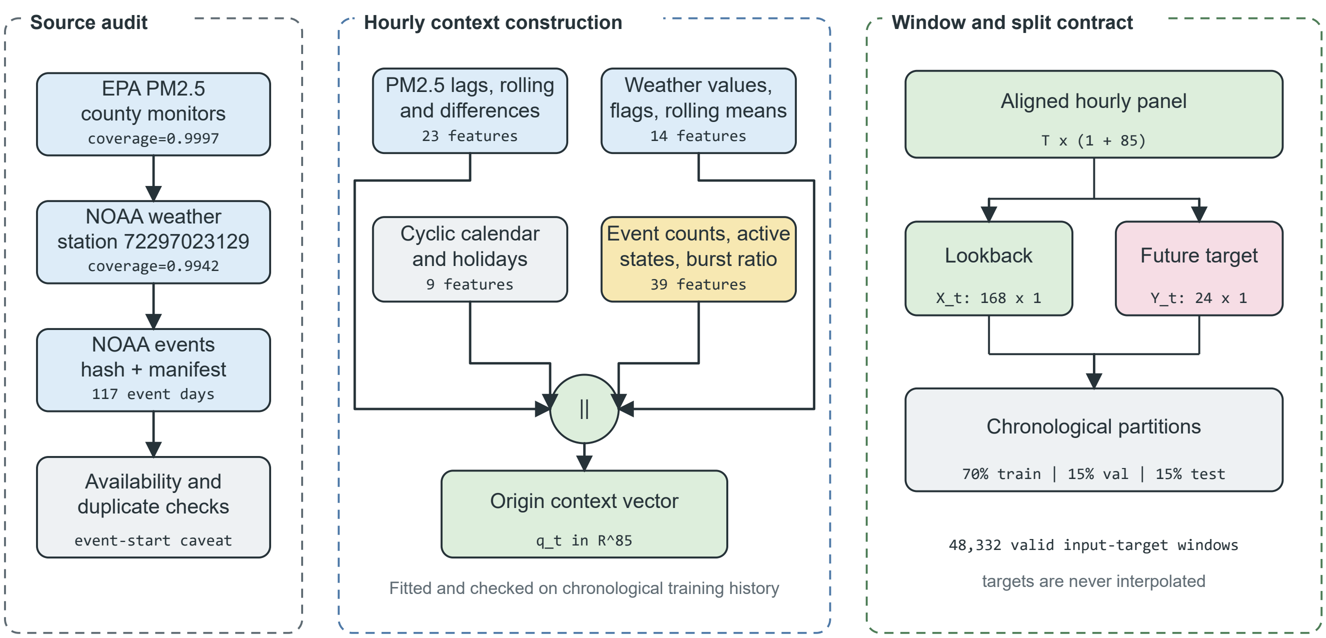

Fig. 2. Source-Audited Context Builder. Source checks feed four feature groups that form an 85-dimensional origin vector. The aligned panel is converted into 168-hour inputs and 24-hour targets before the chronological split.

weather summaries, c_i the list of calendar summaries and e_i the list of event-context summaries available at u_i. The set of temporally eligible origins is defined as in Equation (9) for a query origin t.

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1,\ i \in \mathcal{T}_{train}\}, \tag{9}$$

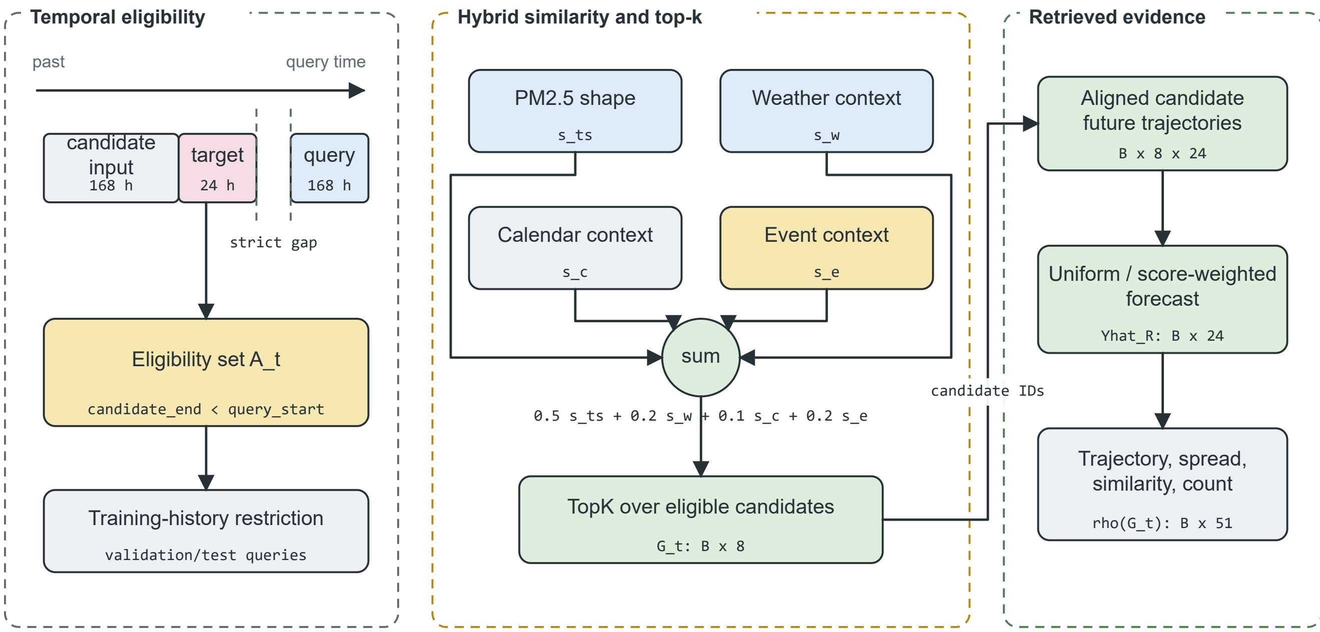

Fig. 3. Leakage-Safe Event-Context Retriever. A strict candidate-target embargo and training-history restriction are applied before hybrid ranking. The top eight candidates provide aligned future trajectories and a 51-dimensional retrieval summary.

To validate and test queries. The strict inequality in (9) means that a candidate's target period ends before the query period is input. The training-history condition for (9) also applies to the final validation and test retrieval. These two schemas, stored and admissible at query time, are defined in Equation (8) and Equation (9), respectively.

The time-series retriever returns the records defined by Equation (10), for a given query window X_t.

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \dots, r_{t,k}^{ts}\}, \tag{10}$$

Here k is the number of retrieved candidates. The default is k=8 and sensitivity analysis of k in $\{1, 4, 8, 16\}$ is performed. Equation (10) is for the records returned before the future trajectory of the records is summarized.

2) Event Knowledge Base: The event knowledge base provides source preserving records, which relate to Los Angeles County. NOAA Hazard Mapping System fire and smoke records are still considered an unmeasured optional extension [36], [37] while NOAA Storm Events are the required structured source. Event identifier is stored, as well as start, end, information-availability assumption, location, category, and source and source URL, title, and summary. Reported records includes wildfire, high wind, heavy rain, flood and other categories of weather; empty categories for missing weather types are set to zero.

Model inputs consist of hourly counts, 24, 72, and 168 hr rolling counts, 72 hr counts broken down into event categories, and active-event counts and an event-burst ratio. Explanation evidence is kept for the last recent source identifiers. The target-horizon overlap is only considered as a post hoc subset label and is not provided as a model input. The publication time is not directly provided by NOAA Storm Events; event start is recorded as the retrospective availability assumption. The records must have real publication or issue dates for strict operational experiments.

3) Hybrid Ranking and Future Alignment: The query and candidate PM2.5 windows are normalized in both their means and standard deviations, and unit length. The candidate statistics for the weather and calendar groups are standardized and mapped to [0,1] by using the cosine similarity. Handle event similarity with clipping to [0,1] and handling of a zero vector. The hybrid retrieval score, which incorporates time-series similarity, weather relevance, calendar relevance and event relevance is given by Equation (11).

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \tag{11}$$

where, s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity and s_e is event relevance. The weights that were used are: $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$ and $\delta = 0.2$. The relevance of each channel of the equation is kept explicit to make it possible to analyze the contribution of each relevance channel without the need for a learned latent scoring function as in (11).

The set to be retrieved is chosen based on Equation (12).

$$G_t = \mathrm{TopK}_{i \in \mathcal{A}_t} \; s(t,i). \tag{12}$$

The only evaluation is done for equation (12) once set $\mathcal{A}_t$ is applied (which is only when $\mathcal{A}_t$ is eligible). The candidate futures are reinstated in the query scale in equation (13).

$$\widetilde{Y}_{t,i} = \mu_t + \sigma_t \left(\frac{Y_i - \mu_i}{\max(\sigma_i, \epsilon)} \right), \qquad i \in G_t, \tag{13}$$

where (μ_t, σ_t) and (μ_i, σ_i) are the statistics of the query and the candidates to look back respectively, and $\epsilon = 10^{-6}$ prevents division by zero. Forecasts for retrieval only are made by taking an average of the aligned candidate futures, uniformly or using non-negative normalized scores. The feature-level path has the same mean trajectory, horizon-wise spread, mean and maximum time-series similarity, and number of candidates, so as to give $\rho(G_t) \in \mathbb{R}^{51}$. The affine transformation in Equation (13) maintains the shape of the candidate trajectory, while reverting the query trajectory's local scale.

D. Drift-Aware Forecast and Evidence Head

The common output layer of two verified forecasting paths and the evidence output is the DFEH. A direct path is defined as the concatenation of the context, retrieval, drift and future-calendar features for 24 independently trained XGBoost regressors, as illustrated in Figure 4. The second path mixes the retrieved forecast with the frozen Chronos-Bolt forecast at a validation-selected weight. Drift scores and flags are retained as diagnostic inputs and evidence fields rather than being used to select between forecasters.

1) Feature-Level Forecasting and Drift Diagnostics: The direct input and prediction to the model is given by Equation (14) for horizon h.

$$u_{t,h} = [q_t \,\|\, \rho(G_t) \,\|\, d_t \,\|\, c_{t+h}], \qquad \hat{y}^X_{t+h} = f_h(u_{t,h}), \quad (14)$$

$u_{t,h} \in \mathbb{R}^{151}$ for the complete M09 model and each f_h is an XGBoost regressor trained with a squared-error objective. The design allows each forecast horizon to have a supervised function while sharing a common origin representation. Equation (14) includes the verified 151 features: 85 context, 51 retrieval, six drift and nine future-calendar values. The drift score and binary flag support regime analysis and the evidence record; they do not adaptively select a forecaster in the reported experiment.

2) Frozen-TSFM Forecast Fusion: The Chronos-Bolt retrieval variant is powered with forecast level fusion, instead of inbuilt Channel Prompting. Define $\widehat{Y}_t^C$ as the "frozen" forecast of Chronos and $\widehat{Y}_t^R$ as the "retrieved" historical forecast. The fused prediction is given by equation (15).

$$\hat{Y}_t^{CR}(\omega) = \omega \hat{Y}_t^C + (1-\omega)\hat{Y}_t^R, \quad (15)$$

where omega in $\{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected omega=0.75. This branch will not change anything on Chronos-Bolt embeddings or weights as Fusion is performed after both forecasts are made. As such, equation (15) is not a prompting operation by the TSFM, rather it is an output-space interpolation.

3) Concept-Drift Evidence: The implementation is distribution shift with transparency-indicators and not claiming that concept drift is identified perfectly. The indicators include the recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift and event-burst ratio. The raw vector that the code works with is specified by equation (16).

$$r_t = \Bigg[|\mu_{24,t} - \mu_{168,t}|, \left| \log \frac{\sigma_{24,t} + \epsilon}{\sigma_{168,t} + \epsilon} \right|, 1 - \mathrm{clip}(\bar{s}_{ts,t}, 0, 1), \sqrt{\frac{1}{4} \sum_{\ell=1}^{4} \left(\frac{w_{t,\ell} - \widetilde{w}_\ell^{tr}}{s_{w,\ell}^{tr}} \right)^2}, \ \max(b_{event,t}, 0) \Bigg], \quad (16)$$

The weather term is based on the temperature, relative humidity, pressure and wind speed, standardized using the values of the training medians and standard deviations. The robust scaling of each component $r_{t,j}$, by the training median and training median absolute deviation (MAD), is done according to equation (17).

$$d_{t,j} = \frac{1}{6} \,\mathrm{clip} \left(\frac{r_{t,j} - \mathrm{med}_{tr}(r_j)}{\max(1.4826 \, \mathrm{MAD}_{tr}(r_j), \epsilon)}, 0, 6 \right). \quad (17)$$

Lastly, the continuous score and binary flag is defined with Equation (18).

$$S_t = \frac{1}{5} \sum_{j=1}^{5} d_{t,j}, \qquad D_t = \mathbb{I}[S_t \geq Q_{0.90}(\{S_v : v \in \mathcal{T}_{val}\})]. \quad (18)$$

Equations (16)–(18) are equivalent to the detector that was used: the reference statistics are set using training data, the threshold 0.90 is set using validation data and test data are transformed without refitting. The binary D_t is retained for diagnostic regime analysis and explanation evidence.

4) Evidence Record and Diagnostic Scale: The explanation module is based on signed local contributions provided by the 24 direct models, weather flags, top three retrieved windows, recent event records, calendar context, forecast direction, and component level drift evidence. The local values are XGBoost TreeSHAP contributions returned by `pred_contribs=True`; the bias column is removed before the values are averaged across horizons. Its diagnostic uncertainty scale is defined by equation (19).

$$u_t^{\mathrm{diag}} = \sqrt{\bar{\sigma}_{R,t}^2 + \overline{\mathrm{RMSE}}_{val}^2}, \quad (19)$$

The scale uses the mean of the 24 horizon-specific validation RMSE values $\overline{\mathrm{RMSE}}_{val}$ and the mean horizon-wise spread of the retrieved trajectories $\bar{\sigma}_{R,t}$. It is not a calibrated prediction interval. Each explanation is constructed from stored fields rather than an unrestricted language model, and every event statement is contextual rather than causal. Equation (19) therefore defines an evidence field and is not used to construct confidence intervals.

E. Relationship to TimeRAF

TimeRAF utilizes a learnable dual encoder with an MLP to retrieve top-k windows from time-series, and uses Channel Prompting [9] to inject candidate embeddings into a pre-trained TSFM. Table II shows which of the principles are kept and which of the mechanisms are changed. It is a methodological comparison not a performance ranking, since the studies are based on different sets of data, horizons and protocols.

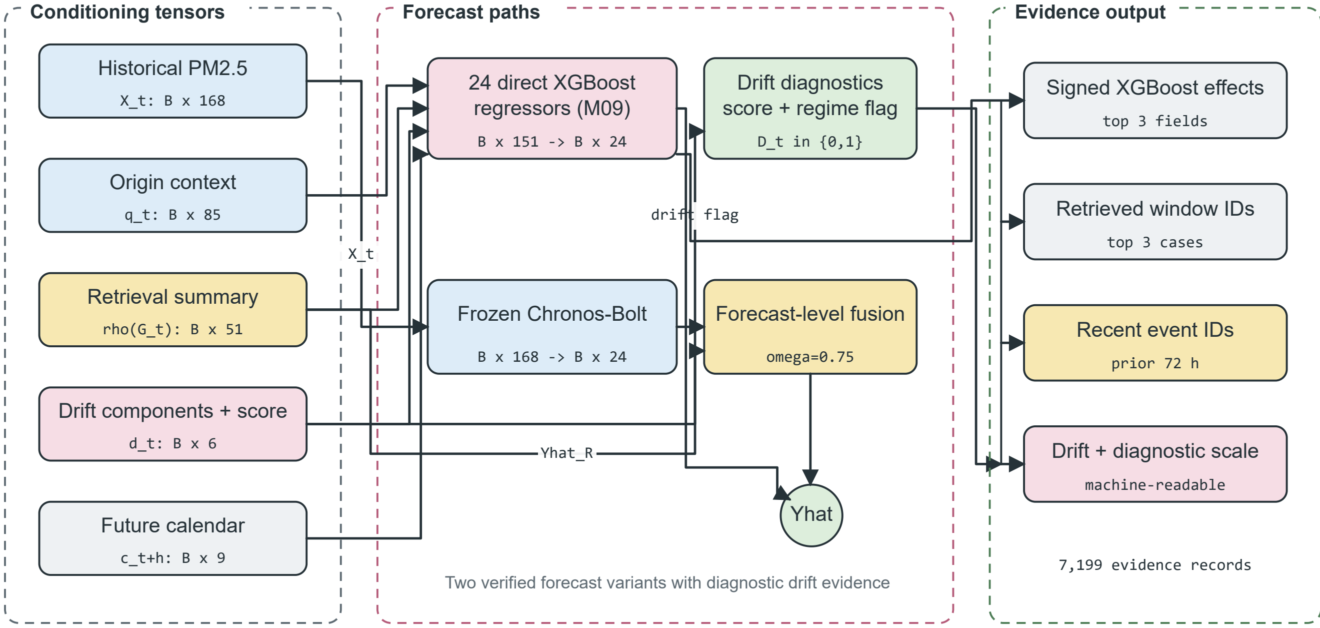

Fig. 4. Drift-Aware Forecast and Evidence Head. Context, retrieval, drift, and future-calendar tensors feed direct horizon regressors; the frozen Chronos-Bolt path is fused only at forecast level. Stored feature effects, retrieved cases, event identifiers, drift components, and uncertainty values form the evidence record.

TABLE II
METHODOLOGICAL COMPARISON OF TIMERAF AND EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF
Primary objective	Improve zero-shot forecasting by augmenting a frozen TSFM with external time-series knowledge.	Evaluate source-audited, event-context retrieval for local PM2.5 forecasting under temporal shift.
Forecasting domain	Multi-domain benchmark datasets.	Los Angeles County hourly PM2.5, 2019–2024.
Knowledge base	Curated, consistently normalized time-series windows.	1,521 training-history windows at the primary 24-hour stride, with denser and sparser sensitivity settings, PM2.5, weather, calendar, and retrospective event context.
Retriever	Learned MLP dual encoder with forecaster-guided training.	Random, cosine, and transparent hybrid similarity; no learned retriever.
Knowledge integration	Internal Channel Prompting of candidate embeddings.	Feature-level XGBoost inputs and forecast-level Chronos-Bolt fusion.
Backbone	Frozen TSFM, principally TTM-Base.	Direct XGBoost horizon models and frozen `amazon/chronos-bolt-small`.
Leakage control	Non-overlapping knowledge windows and cross-dataset training constraints.	Candidate windows may overlap one another, but every candidate target must end before the query lookback; validation and test use training history only.
Event handling	Not an explicit module.	Availability-tagged NOAA Storm Events features and source-record evidence, reported retrospectively.
Drift and explanation	Retrieved-case inspection.	Five-component drift audit, signed feature effects, retrieved IDs, event IDs, and diagnostic scale.
Implementation scope	Learned retriever and Channel Prompting are implemented.	Those mechanisms are not implemented and remain future extensions.

VII. EXPERIMENTAL SETUP

This section describes the final experiment. Model-selection choices were made from the training and validation partitions before test evaluation. Run identifier `20260810T103436161252Z` was used to verify artifact consistency.

A. Dataset Description and Audit

The study period is 2019-2024. Los Angeles County PM2.5 data were collected from US EPA, AirData/AQS parameter 88101 [34]. The official Los Angeles County AQS monitors' hourly county median was used as the audited target series (not any one monitor site). This fall back picked 06-037-COUNTY and observed target hours of 52,602 and observed PM2.5 coverage of 0.9997. The PM2.5 value was always measured in micrograms per cubic meter.

TABLE III
DATA-READINESS AUDIT

Gate	Observed	Requirement	Status
PM2.5 coverage	0.9997	≥ 0.70	Pass
Weather coverage	0.9942	≥ 0.95	Pass
Event days	117	≥ 30	Pass
Source-overlap days	2,192	≥ 180	Pass
Qualifying categories	3	≥ 2	Pass
Strict availability	False	—	Caveat

Weather covariates data was downloaded from NOAA Global Hourly/ISD, using the NOAA Open Data Dissemination public buckets [35]. The selected station was 72297023129 (Long Beach / Daugherty Field / Airport) at a distance of 10.28 km to the PM2.5 target centroid. The full weather-feature coverage following past-only short-gap filling was 0.9942 and the raw measured core coverage was 0.9905.

NOAA Storm Events annual detail archives [36] were extracted and loaded as environmental events, with duplicate events removed. The experiment used a locally staged cache of unchanged NOAA annual archives and a source manifest listing the official NOAA URLs, byte count and SHA-256 hash for each archive. No delivery-mode change was required during the audit, and the `verified_official_noaa_cache` mode was retained. The event audit identified 117 event days, 2,192 source-overlap days and three qualifying event categories. High wind (76 records), flood (61), and other weather (40) meet the configured 20-record category threshold; heavy rain (18) and wildfire (9) are retained but do not qualify. Because NOAA Storm Events does not provide machine-readable publication timestamps, the event-aware outputs retrospectively assume availability at the event start time and are not, by themselves, real-time event forecasts. The resulting readiness status is given in Table III. Figure 2 shows the source audit window and window contract.

Window construction started from 52,608 aligned hourly rows and 52,417 possible forecast origins. It excluded 48 origins at split boundaries, eight for a missing lookback, 144 for a missing target, and 3,885 for missing origin features; none was excluded for future-calendar values. The remaining 48,332 origins comprise 34,020 training, 7,113 validation, and 7,199 test origins.

Table IV summarizes the assessed scope and the various sources' contribution to the forecasting process.

B. Data and Code Availability

EPA AirData, NOAA Global Hourly/ISD and NOAA Storm Events provide the raw public records and download pages [34]–[36]. Because the local execution environment could not access NCEI directly, the experiment used a locally staged NOAA Storm Events cache, and every cached NOAA file was checked against the generated source manifest. Source code is available at https://github.com/shasabbir/event-time-raf. The implementation uses Python 3.12.13, NumPy, pandas, scikit-learn, XGBoost, PyTorch and the chronos-forecasting package. The repository contains the configuration, notebooks, source modules and tests required to reproduce the pipeline when the public source files are available.

C. Forecasting Task and Split

The forecasting task involves looking at historical information for a 168-hour period and predicting PM2.5 for the next 1-24 hours. The split is done in chronological order, the first 70% of valid windows are for training, the next 15% of valid windows for validation and the last 15% of valid windows for testing. The final test set has 7,199 forecast origins and 172,776 forecast target points. Test origins span 7 February through 30 December 2024 UTC, and their targets end on 31 December 2024 UTC. Random splitting is NOT performed.

D. Models

The evaluation includes persistence and seasonal baselines, hourly-by-month climatology, ridge regression, direct XGBoost models, random, cosine, calendar-only and event-stratified retrieval, Event-TimeRAF with and without drift indicators, frozen Chronos-Bolt, and three validation-tuned Chronos fusion variants. The fusion controls replace retrieval with climatology or persistence while keeping the same weight-selection process. Auxiliary A00 removes event features and A01 uses random retrieval so that event and retrieval effects can be separated. The foundation-model checkpoint is `amazon/chronos-bolt-small`.

E. Implementation and Hyperparameters

This experiment was conducted in a cloud-based Linux 6.12.90 environment with Python 3.12.13. The runtime was 3,319.95 seconds. Package versions recorded in the manifest include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128 and chronos-forecasting 2.3.1. The main hyperparameters are given in Table VI.

F. Evaluation Metrics and Statistical Checks

The main metrics are: MSE, MAE, RMSE, MAPE, sMAPE and R^2. Although MSE is still used for comparison with the TimeRAF evaluation, the use of MAE and RMSE is for an interpretable scale of the PM2.5 error. The following equation (20) shows the key error measures for n observations y_j of the forecast and prediction $\hat{y}_j$.

$$\begin{aligned} \mathrm{MSE} &= \frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{j=1}^{n}|y_j - \hat{y}_j|, \\ \mathrm{RMSE} &= \sqrt{\mathrm{MSE}}, \end{aligned} \tag{20}$$

The Equation (21) is the summary of explained variance.

$$R^2 = 1 - \frac{\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}{\sum_{j=1}^{n}(y_j - \bar{y})^2}. \tag{21}$$

TABLE IV
DATASET AND SOURCE SUMMARY

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2024; 52,602 observed target hours	Target sequence and lag/rolling features
Weather	NOAA Global Hourly/ISD, station 72297023129	2019–2024; 0.9942 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	204 Los Angeles records; 117 event days	Retrospective event-start features and explanation evidence
Calendar	Generated from timestamps and public holiday rules	2019–2024 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE V
EVALUATED MODEL FAMILIES

ID	Description
M00–M02	Persistence and seasonal naive baselines
M03–M04	XGBoost PM2.5-only and context models
M05–M06	Random and cosine retrieval-only baselines
M07	XGBoost with cosine retrieval features
M08–M09	Event-TimeRAF without/with drift indicators
M10–M11	Frozen Chronos-Bolt and retrieval fusion
C00–C03	Climatology, ridge, calendar, and event-stratified controls
C10–C11	Chronos climatology and persistence fusion controls

The same forecast origins and horizons are used for all overall metrics. MAPE and sMAPE remain in machine-readable outputs, but are not used for ranking because low PM2.5 values can make percentage errors unstable. Subset metrics are reported when at least 50 origins are available. The event, active-event, and drift subsets contain 532, 293, and 566 origins. Primary comparisons use 2,000 paired block-bootstrap resamples with 168-origin blocks. Diebold–Mariano tests use a HAC lag of 167, followed by Holm adjustment within each pre-specified metric family.

G. Comparison with Existing Studies

The direct comparison of these results with other TSFM and air-quality papers is not valid as the papers employ different sets of data, forecasting periods, variables, and evaluation schemes. For this reason, the methodological coverage is compared, in Table VII, and not cross-paper performance dominance claimed.

VIII. RESULTS AND DISCUSSION

A. Overall Comparison with Baselines

Table VIII gives the test results. The strongest model is M04, which uses PM2.5 history, weather, and calendar features. It achieves MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379. Full Event-TimeRAF M09 obtains MSE 26.734 and MAE 3.144. Therefore, adding retrieval, event, and drift features does not improve the all-origin result over M04.

Validation selected a retrieval-fusion weight of 0.75. M11 changes frozen Chronos-Bolt MSE from 28.941 to 28.733, but the M11–M10 interval is $[-0.654, 0.216]$ and the Holm-adjusted DM p-value is 1.000. This difference is unresolved. The matched climatology fusion C10 obtains MSE 27.450 and is better than M11 by 1.283 MSE, with interval $[0.694, 1.913]$ and adjusted $p < 0.001$. Persistence fusion selects a TSFM weight of 1.0 and is identical to M10. The observed fusion result is therefore not retrieval-specific.

B. Ablation Study

Table IX compares the main components. Weather and calendar context lowers MSE by 3.764 compared with PM2.5-only XGBoost, and cosine retrieval is much better than random retrieval when retrieval is used alone. Other feature-level additions have intervals that cross zero. M09 is 0.550 MSE worse than M04, its difference from event-free A00 is unresolved, and the drift indicators change M08 by only 0.034 MSE.

The intervals are paired block-bootstrap intervals. Holm-adjusted DM values are shown only for the pre-specified primary family; dashes indicate secondary diagnostic comparisons.

The cosine retrieval baseline improves as k increases in the tested range. At $k = 16$, MSE is 34.440. The feature models retain the pre-specified default $k = 8$.

C. Knowledge-Base Density and Event Sensitivity

The size of the historical memory matters for retrieval. Table XI shows that a six-hour stride gives the best retrieval-only result. It increases the candidate count from 191 at the 192-hour stride to 5,644 and reduces MSE from 39.037 to 34.616. A one-hour stride does not improve it further. The full feature model also has its lowest observed MSE at six hours, 26.412, but it is still behind M04 at 26.185.

The event channel changes the selected candidates, but does not improve their forecasts. With event weight 0.2, at least one top-eight candidate changes for 91.4% of test queries and the event-candidate share for event-context queries increases from 32.9% to 95.7%. Even so, Table XII shows higher all-origin and event-subset error as the event weight rises. This result applies to the tested count-based representation and ranking rule; it does not show that environmental events have no forecast signal.

D. Event and Drift Subsets

Table XIII gives event and drift subset results. Event targets have variance 22.636 compared with 43.346 for non-event targets. Drift and non-drift target variances are 69.593 and 39.826. Raw subset MSE therefore reflects both forecasting

TABLE VI
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

Component	Setting
Study period	2019–2024
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	70% train, 15% validation, 15% test, chronological
SACB feature groups	23 PM2.5, 14 weather, 9 calendar, 39 event features
SACB missing-data rule	Past-only filling for gaps up to 3 hours; no target interpolation
LSER retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
LSER knowledge base	1,521 windows at 24-hour primary stride; sensitivity at 192, 24, 6, and 1 hours
LSER similarity weights	time series 0.5, weather 0.2, calendar 0.1, event 0.2
LSER event-weight sensitivity	$\delta \in \{0, 0.2, 0.5, 0.8, 1.0\}$ with remaining weights renormalized
DFEH direct input	151 features for each of 24 horizon regressors
DFEH XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
DFEH drift detector	5 components; 24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
DFEH Chronos path	`amazon/chronos-bolt-small`, batch size 64
DFEH fusion weights	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected $\omega = 0.75$
Explanation record	Top 3 feature effects, windows, and recent event IDs
Inference	2,000 paired block-bootstrap resamples, 168-origin blocks; DM HAC lag 167; Holm adjustment

TABLE VII
METHODOLOGICAL COMPARISON WITH RELATED FORECASTING WORK

Method family	TSFM	Retrieval	Events	Explainability	Comment
PatchTST-style Transformers [22]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-aware PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
Event-TimeRAF	Yes	Yes	Yes	Evidence-grounded	Adds official event audit, temporal leakage control, drift indicators, and Chronos retrieval fusion.

TABLE VIII
MANIFEST-BACKED OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
C00 hour–month climatology	36.297	4.038	6.025	0.139
C01 ridge context	28.448	3.226	5.334	0.325
C02 calendar retrieval	45.534	4.429	6.748	-0.080
C03 event-stratified retrieval	38.428	3.924	6.199	0.089
M00 persistence	47.649	4.137	6.903	-0.130
M01 daily seasonal	48.706	4.057	6.979	-0.155
M02 weekly seasonal	65.606	5.123	8.100	-0.556
M03 XGBoost PM2.5	29.949	3.368	5.473	0.290
M04 XGBoost context	**26.185**	**3.125**	**5.117**	**0.379**
M05 random retrieval	44.089	4.346	6.640	-0.046
M06 cosine retrieval	36.243	3.818	6.020	0.140
M07 XGBoost cosine	26.442	3.132	5.142	0.373
M08 Event-TimeRAF no drift	26.700	3.144	5.167	0.367
M09 Event-TimeRAF full	26.734	3.144	5.171	0.366
M10 frozen Chronos-Bolt	28.941	3.205	5.380	0.314
M11 Chronos + retrieval	28.733	3.205	5.360	0.319
C10 Chronos + climatology	27.450	3.137	5.239	0.349
C11 Chronos + persistence	28.941	3.205	5.380	0.314

skill and target difficulty. The drift subset is only used for stratified evaluation.

E. Horizon-Wise and Retrieval Diagnostics

Figure 5 and Fig. 6 show the horizon-wise test errors. Retrieval behavior is summarized in Fig. 7, and Fig. 8 shows the drift scores. Figure 9 gives one representative forecast. All plots are generated from run `20260810T103436161252Z`. The upper-tail drift threshold flags 10.0% of validation origins and 7.9% of test origins. This is a diagnostic result only; the flag does not select a model or prove that concept drift occurred.

F. Evidence and Traceability Analysis

Within the DFEH, traceability takes the form of a source-grounded evidence record, not as a post-hoc, free-text in-

TABLE IX
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

Comparison	MSE diff.	95% CI low	95% CI high	Holm DM p
M04 minus M03, weather/calendar	-3.764	-7.120	-1.709	0.067
M06 minus M05, cosine/random retrieval	-7.846	-11.333	-4.849	—
M07 minus M04, retrieval summary	0.258	-0.269	1.012	—
M08 minus M07, event-hybrid ranking	0.258	-0.087	0.690	—
M09 minus M08, drift indicators	0.034	-0.079	0.147	—
M09 minus A00, event context	0.216	-0.114	0.635	1.000
M09 minus M04, complete feature path	0.550	-0.122	1.592	1.000
M11 minus M10, retrieval fusion	-0.209	-0.654	0.216	1.000
M11 minus C10, climatology placebo	1.283	0.694	1.913	< 0.001

TABLE X
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	60.555	4.886	7.782
4	40.004	4.010	6.325
8	36.243	3.818	6.020
16	34.440	3.711	5.869

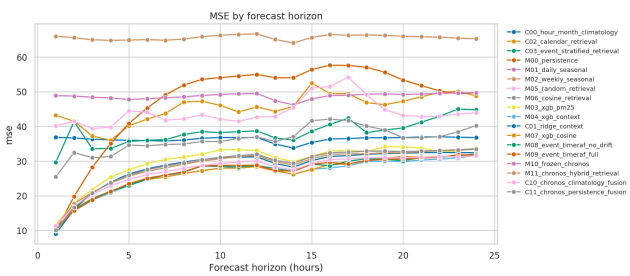

Fig. 5. MSE by forecast horizon on the Los Angeles County test set.

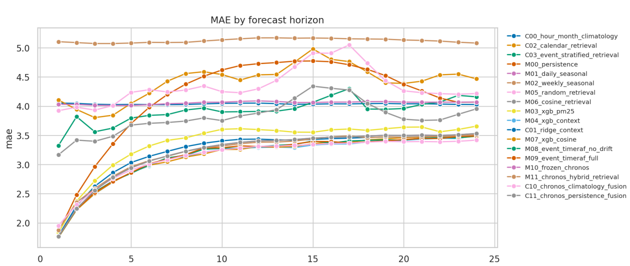

Fig. 6. MAE by forecast horizon on the Los Angeles County test set.

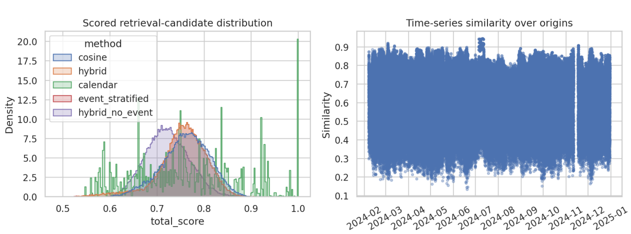

Fig. 7. Similarity and candidate diagnostics for test-set retrieval.

terpretation. The following information is stored in the pipeline for each forecast origin evaluated: Run ID, Window ID, Origin Timestamp, Retrieved Historical Evidence IDs,

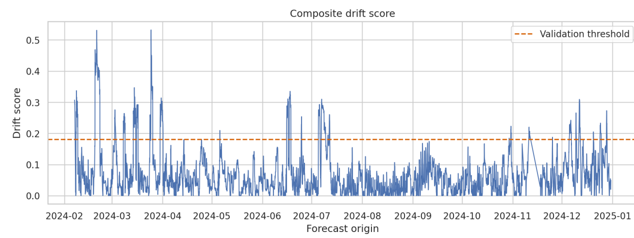

Fig. 8. Composite drift-score diagnostics on the test set.

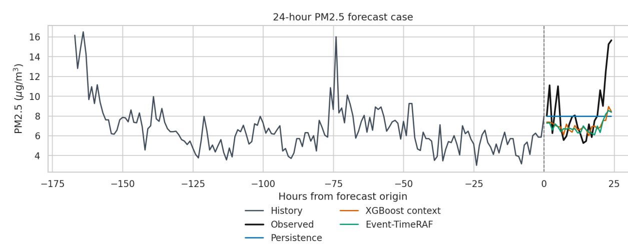

Fig. 9. Representative 24-hour forecast case from the test set.

Event Evidence IDs, Leading Feature Effects, Drift Components, Retrieval Spread, Validation Residual RMSE, Diagnostic Uncertainty Scale, Drift Score, Drift Flag, and Event-availability mode. The result of the evaluation is contained in `explanations.parquet` (7,199 explanation records). A representative record contains the strongest local contribution fields (the current PM2.5 value and recent rolling PM2.5 means), links the prediction to three retrieved historical windows, links recent source-recorded storm events when available, and marks distribution shift when the drift detector is active.

The summary of the evidence layer is given in Table XIV. This design supports auditing because each statement is mapped to a model feature, a retrieved time-series window, an official event record or a drift component. Event records are contextual evidence rather than causal explanations, and the drift flag is an uncertainty warning. Chronos-Bolt's nominal 80% interval covers 78.2% of test targets and has mean width 9.890 μg/m^3. This interval is separate from the diagnostic scale, which is not calibrated.

G. Computational Cost and Reproducibility

The entire experiment ran in a cloud-based Linux environment with Python 3.12.13 in 3,319.95 seconds. Recorded pack-

TABLE XI
KNOWLEDGE-BASE STRIDE SENSITIVITY ON THE TEST SET

Stride (h)	Candidates	Retrieval MSE	Retrieval MAE	Full M09 MSE
192	191	39.037	3.991	26.712
24	1,521	38.411	3.921	26.734
6	5,644	**34.616**	**3.660**	**26.412**
1	34,020	36.416	3.790	26.740

TABLE XII
EVENT-CHANNEL SENSITIVITY FOR RETRIEVAL-ONLY FORECASTS

Method	Event weight	Top-k changed	All-origin MSE	Event-subset MSE
Hybrid	0.0	0.000	**38.080**	**14.299**
Hybrid	0.2	0.914	38.411	16.401
Hybrid	0.5	0.914	38.703	19.292
Hybrid	0.8	0.914	38.969	21.201
Hybrid	1.0	1.000	40.860	27.130
Event-stratified	0.2	0.917	38.428	16.638

TABLE XIII
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

Subset	Model	Origins	MSE	MAE
Non-event	M04 XGBoost context	6,667	27.520	3.179
Event	M04 XGBoost context	532	9.451	2.450
Non-event	M09 Event-TimeRAF full	6,667	27.995	3.187
Event	M09 Event-TimeRAF full	532	10.932	2.613
Non-drift	M04 XGBoost context	6,633	26.060	3.086
Drift	M04 XGBoost context	566	27.650	3.585
Non-drift	M09 Event-TimeRAF full	6,633	26.282	3.085
Drift	M09 Event-TimeRAF full	566	32.039	3.833
Drift	M10 frozen Chronos-Bolt	566	29.539	3.524
Drift	M11 Chronos + retrieval	566	29.210	3.610

TABLE XIV
EXPLANATION EVIDENCE STORED BY EVENT-TIMERAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Diagnostic scale	Combined retrieval spread and validation residual RMSE; not a calibrated interval.

TABLE XV
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

Item	Recorded value
Run identifier	`20260810T103436161252Z`
Runtime	3,319.95 seconds
Platform	Cloud-hosted Linux environment
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	2,000; block length 168 origins
Chronos checkpoint	`amazon/chronos-bolt-small`
Code availability	https://github.com/shasabbir/event-time-raf

age versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and chronos-forecasting 2.3.1. The run manifest contains the configuration hash and artifact checksums. CPU, GPU, RAM, and peak-memory identifiers were not recorded, so the runtime is not presented as a hardware-normalized efficiency result.

The overall results show that supervised context features are strong for this task. Dense memory improves retrieval alone, but retrieval, event, and drift features do not beat M04. The M11 change from frozen Chronos is unresolved, and the climatology fusion control is significantly better than M11. Event-TimeRAF is therefore an auditable evaluation framework with a scoped negative result, not evidence of retrieval-specific or event-specific superiority.

Lastly all event aware interpretations should be read with the caveat of availability. NOAA Storm Events records are official and hash verified, and the source doesn't support getting the publication time in machine readable format. Event-related results are retrospective sensitivity results as the experiment measures the availability time as the start time for the event. The true issue or publication times of events or smoke sources would need to be used in a strict operational study, e.g. an audited NOAA HMS cache or alert product archive.

IX. LIMITATIONS

The first is the availability of the events. NOAA Storm Events records are official and are hash verified, and the public detail files don't contain machine-readable publication time. The SACB therefore uses event start as an availability proxy and the results are retrospective sensitivity analyses and not operational forecasts.

The geographic validation is the second constraint. The experiment covers one county-level PM2.5 aggregate for the period 2019-2024. It does not entail transfer to another county, monitoring topology, climate and emissions regime and thus not meet the external validation criterion for a mature generalization claim.

The third limitation is related to integration of models. The LSER is not based on a learned retriever but instead on transparent similarity functions and the DFEH is a combination of Chronos-Bolt and retrieval at forecast level. Event-TimeRAF consequently does not duplicate the learned dual encoder or built-in Channel Prompting of TimeRAF.

The fourth restriction concerns computational reporting. No hardware model identifier or peak-memory value was reported in the run manifest; therefore, runtime cannot be interpreted as a hardware-normalized efficiency value.

The fifth limitation is target aggregation. The hourly county median combines a changing monitor set and suppresses local extremes. This improves coverage, but can weaken event relationships and does not support neighborhood- or monitor-level conclusions.

X. Future Work

Operational event studies should replace event-start proxies with archives that reveal the true issue or publication time and should test sources more directly related to PM2.5, such as smoke observations. External validation should apply the same protocol to a second US county. Future work can assess site-level targets, a learned event-context retriever, richer event embeddings, calibrated uncertainty, and internal TSFM event conditioning. These extensions should retain the source audit, temporal embargo, and component controls used here.

XI. Conclusion

This study audited event-aware retrieval for 24-hour PM2.5 forecasting in Los Angeles County. The pipeline aligns official records, applies a strict query–candidate embargo, tests memory density and event sensitivity, and keeps traceable evidence. M04 is the strongest model, with MSE 26.185, MAE 3.125, RMSE 5.117, and R^2 0.379. Full M09 obtains MSE 26.734. M11 changes frozen Chronos MSE from 28.941 to 28.733, but its paired interval crosses zero. Climatology fusion reaches 27.450 and significantly outperforms M11. Dense retrieval improves retrieval alone, while the tested event channel does not improve it. The main contribution is a reproducible evaluation and a bounded negative result: retrieval can be implemented under a verified query–candidate temporal embargo and audited carefully without necessarily improving on strong local context or simple forecast combination.

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